NSE Outlier Detective

System Overview Document

AI-Powered Financial Outlier Detection
for NSE-Listed Companies

Prepared for: Management & Stakeholders
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1. What Is This System?

The NSE Outlier Detective is an AI-powered financial analysis dashboard that automatically identifies companies in any sector of the National Stock Exchange (NSE) that behave significantly different from their peers - either positively or negatively.

Think of it as an automated financial analyst that:

- Pulls real-time financial data for all companies in a chosen sector
- Computes 30+ financial ratios for each company
- Compares every company against its sector peers
- Flags the ones that stand out as statistical outliers
- Explains why they stand out and whether it's a risk or an opportunity

NOTE: The system does not provide investment advice. It highlights anomalies that warrant further human investigation - saving analysts hours of manual spreadsheet work.

2. How Does It Work?

The system follows a four-stage pipeline:



2.1 Data Collection

The system connects to Yahoo Finance (yfinance) and automatically pulls the latest financial statements for every company in the selected sector:

Statement	Period	Purpose
Income Statement	Last 2 years (quarterly)	Revenue, margins, profit trends
Balance Sheet	Last 2 years (quarterly)	Debt levels, asset quality, liquidity
Cash Flow Statement	Last 2 years (quarterly)	Cash generation, capital expenditure
Market Data	Current	P/E, P/B, market cap, enterprise value

2.2 Ratio Computation

The system calculates 30+ financial ratios, organized into five pillars:

Pillar	Key Ratios	What It Measures
Profitability	Gross/Operating/Net Margin, ROE, ROA	How efficiently profit is generated
Liquidity	Current, Quick, Cash Ratio	Can short-term obligations be met?
Solvency	Debt/Equity, Debt Ratio, Interest Cov.	How much debt risk is carried?
Efficiency	Asset/Inventory/Receivables Turnover	How well resources are used
Growth	Revenue/Income/EBITDA/OCF Growth	Is the company growing?

2.3 Outlier Detection

Three independent statistical methods identify outliers:

Method	How It Works	What It Catches
IQR Fencing	Flags values outside 1.5x IQR	Values far from sector middle 50%
Z-Score	Flags >2.5 std deviations from mean	Extreme statistical deviations
Isolation Forest	ML algorithm for multi-dim anomalies	Multi-ratio anomalies

Each outlier receives a severity score (0-5) and is classified as either a Positive Outlier (unusually strong performance) or a Negative Outlier (red flag).

2.4 Insights & Reporting

The system generates plain-English explanations using rule-based logic, identifying risk flags (high debt, declining margins) and opportunity signals (fortress balance sheets, superior growth). Results can be exported as Excel workbooks or CSV files.

3. The Dashboard - What Each Tab Does

The dashboard has 8 tabs, each serving a specific analytical purpose.

3.1 Home - Landing Page

The entry point of the application. Provides an overview of the system's capabilities and guides the user to select a sector and begin analysis.

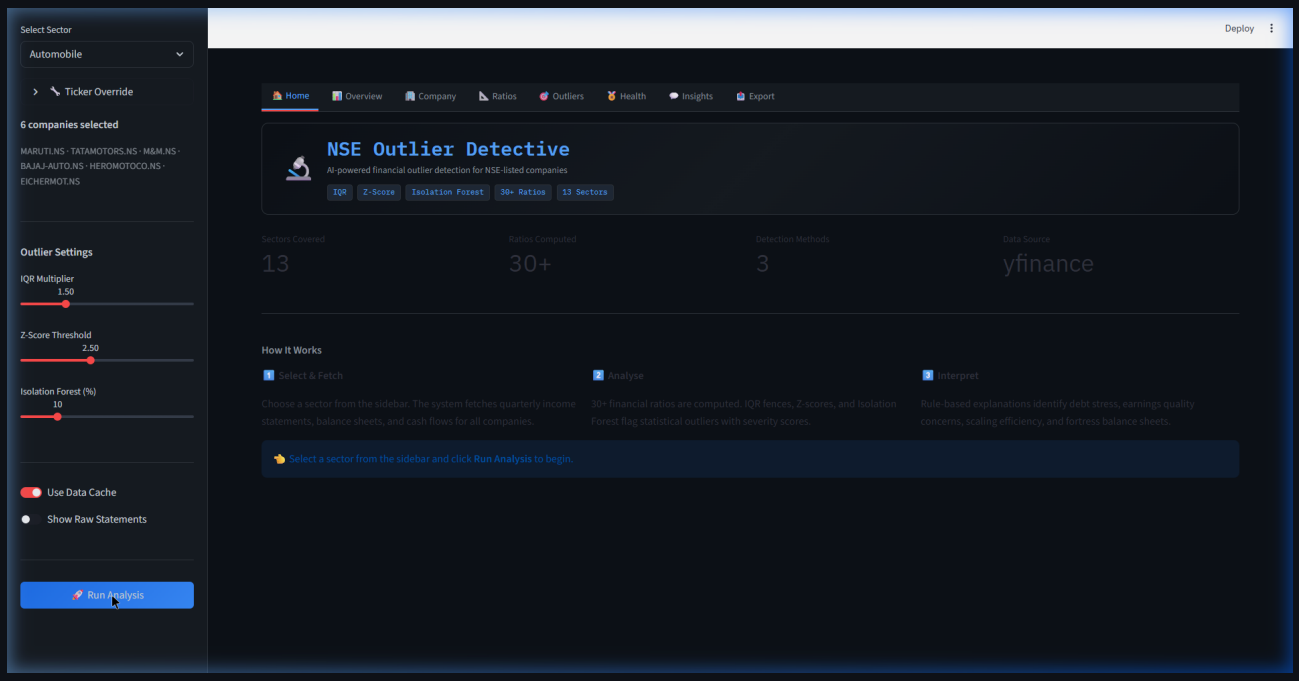


Figure 1 - Home Page

3.2 Overview - Sector-Wide Dashboard

After running an analysis, this tab shows key metrics (companies analysed, total outliers), a color-coded Peer Ratio Matrix comparing all companies side-by-side, and a Z-Score normalized heatmap. The coloring is direction-aware: green always means 'financially healthy' regardless of whether higher or lower values are better for that ratio.

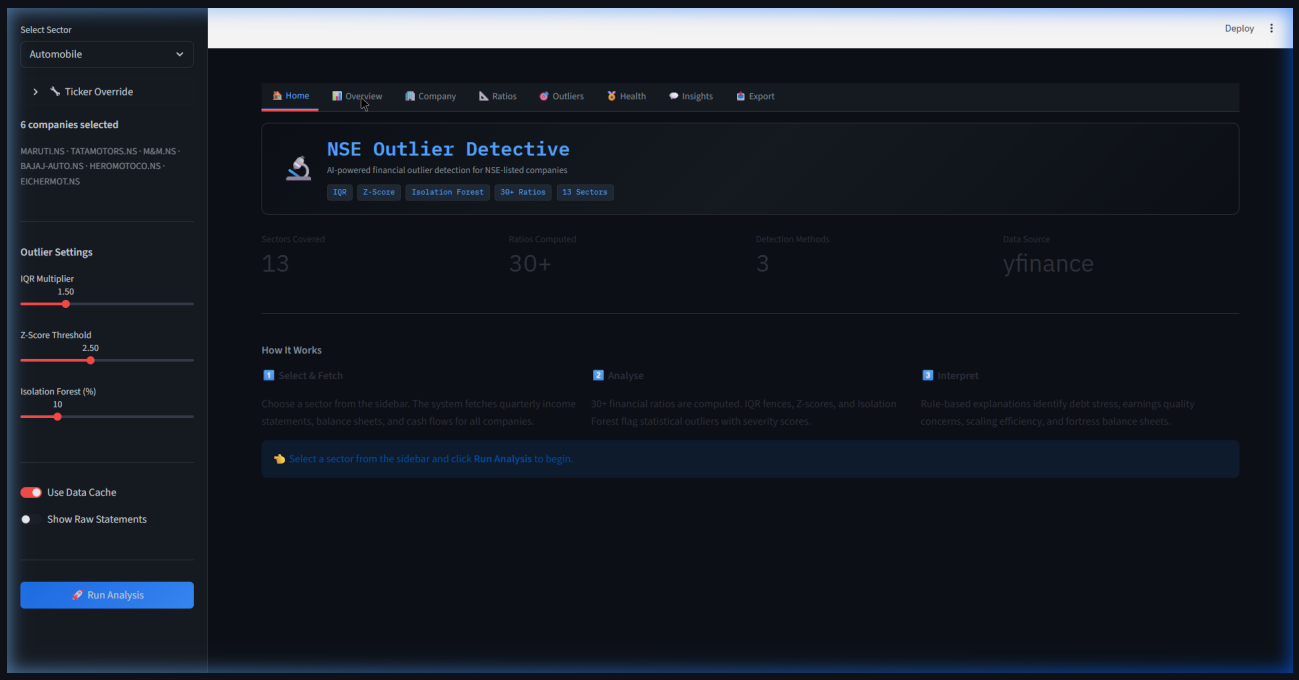


Figure 2 - Overview Tab with Peer Ratio Matrix

3.3 Company - Individual Deep Dive

Select any company to see its market data (price, market cap, P/E), financial ratio table with sector percentile ranks, a radar chart showing performance across all five pillars, and specific outlier flags for that company.

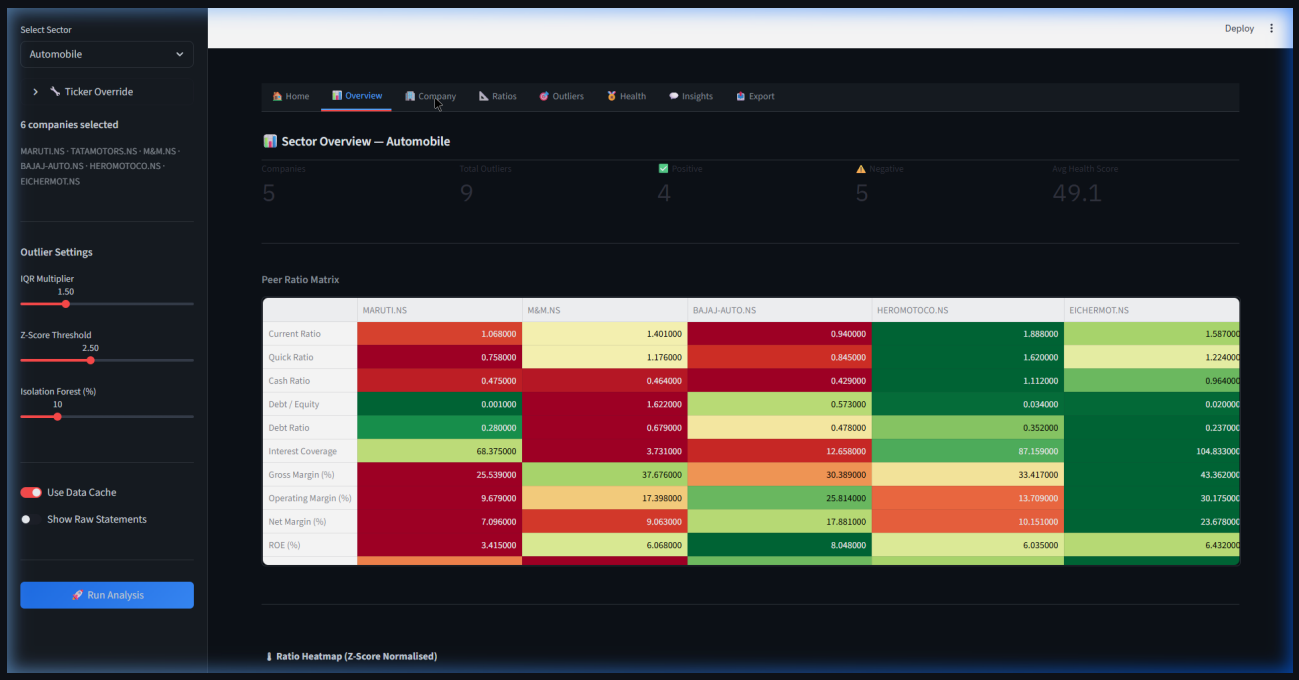


Figure 3 - Company Tab with Radar Chart

3.4 Ratios - Detailed Ratio Analysis

Three sub-tabs: Distribution (box plots showing ratio spread across the sector), Correlation (heatmap showing inter-ratio relationships), and Peer Comparison (full table of actual ratio values with direction-aware coloring).

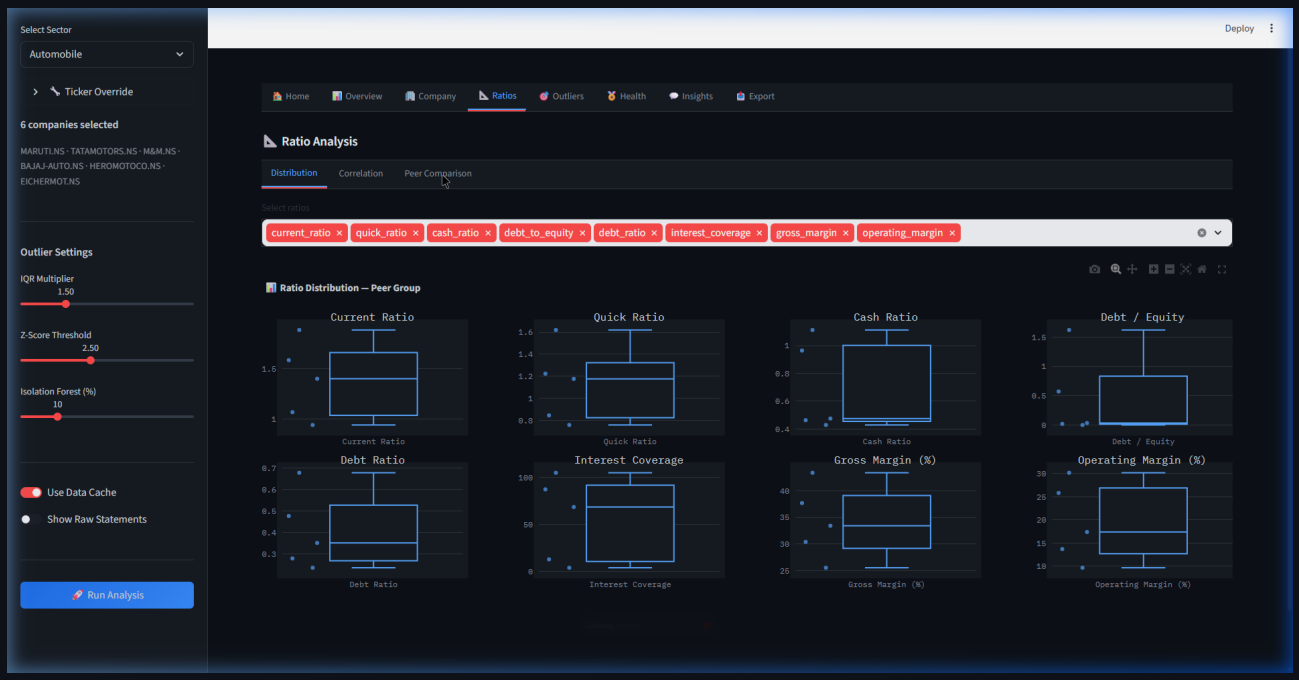


Figure 4 - Ratios Tab: Peer Comparison

3.5 Outliers - Detection Dashboard

The core analytical screen. Features interactive filters (All/Positive/Negative, minimum severity, specific ratio), a scatter plot mapping Deviation vs. Severity, a sortable Top Outliers table, box plots with outlier companies highlighted, Isolation Forest anomaly scores, and one-click CSV export of all outlier records.

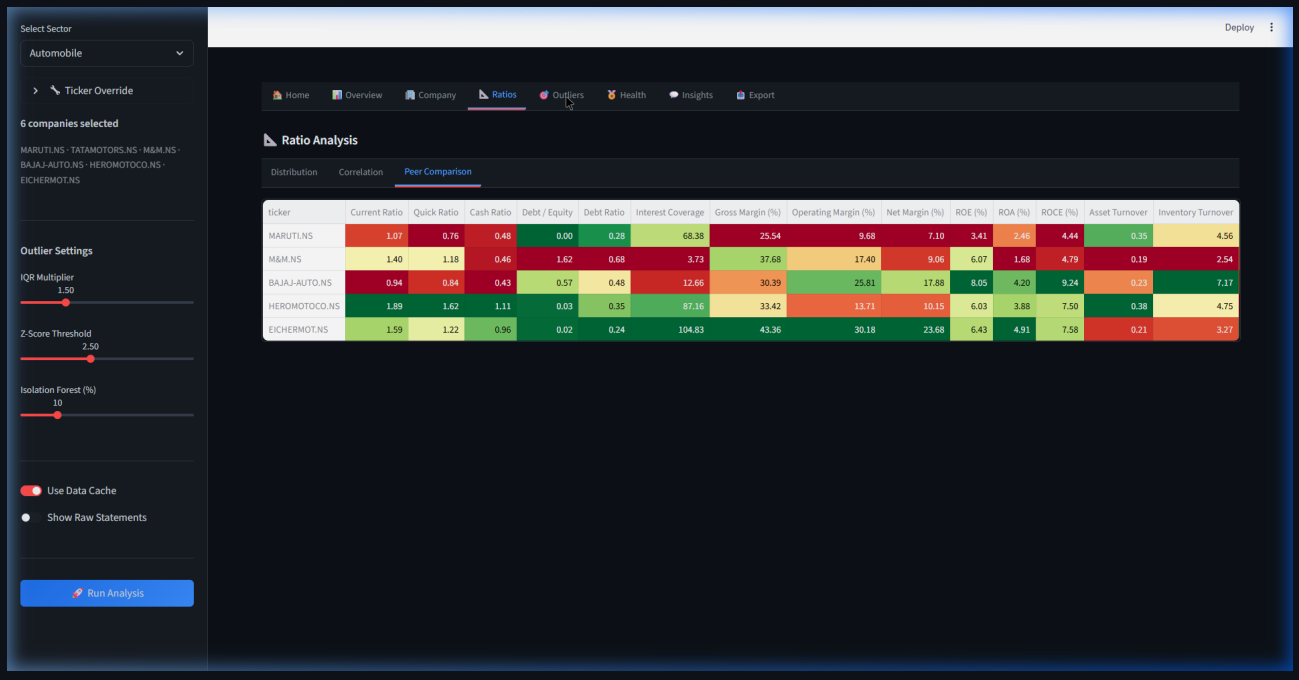


Figure 5 - Outliers Tab: Scatter Plot & Top Outliers Table

3.6 Health - Financial Health Scorecard

Each company receives a composite Financial Health Score (0-100) based on weighted pillar scores. Shows a ranking chart, pillar score breakdown table, and radar comparison.

Pillar	Weight
Profitability	25%
Liquidity	20%
Solvency	25%
Efficiency	15%
Growth	15%

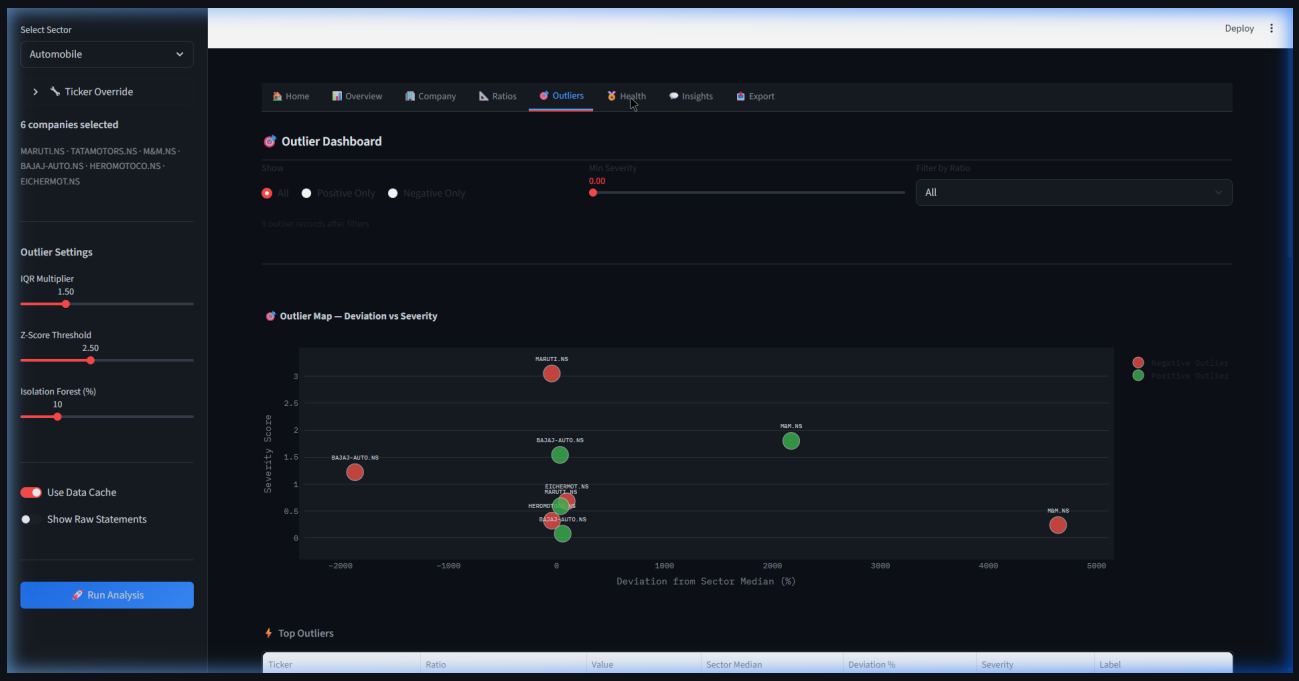


Figure 6 - Health Tab: Company Rankings

3.7 Insights - AI-Generated Explanations

Automatically generated plain-English insights for each company, organized into Risk Signals and Opportunity Signals. Example: 'M&M.NS has a Debt-to-Equity ratio of 1.62, significantly above the sector median of 0.03 - indicating higher leverage risk.'

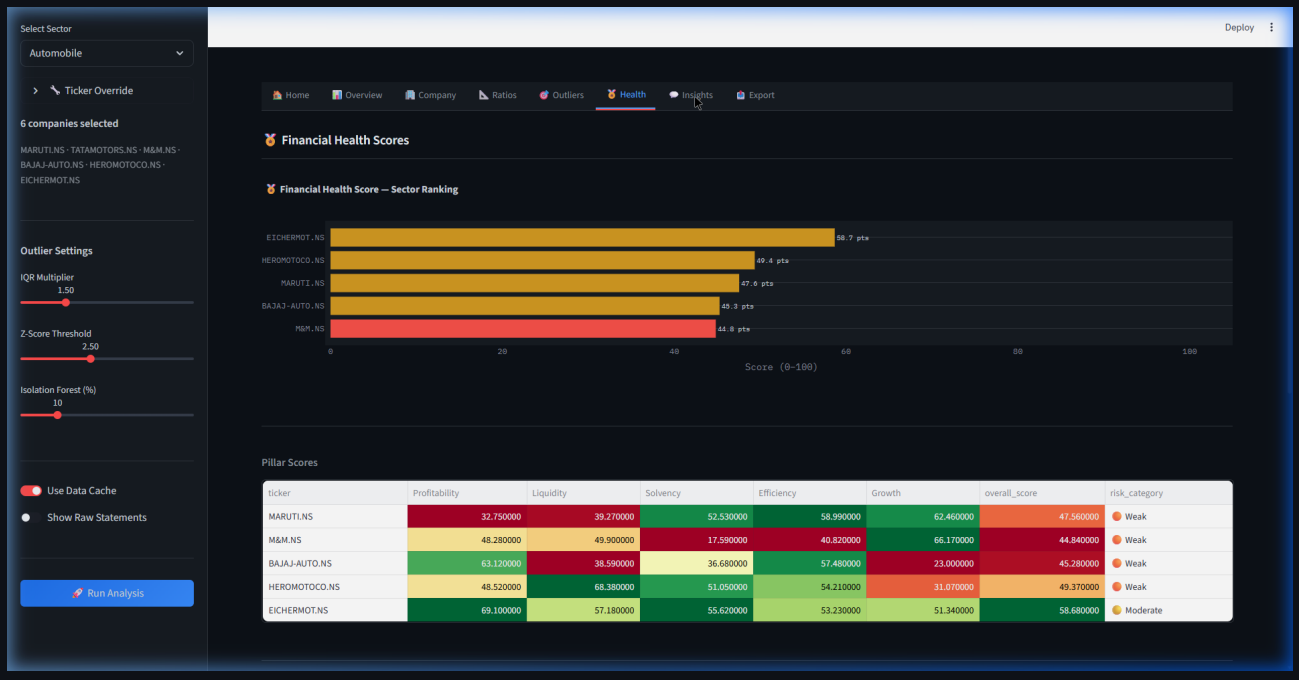


Figure 7 - Insights Tab: Risk & Opportunity Signals

3.8 Export - Download Reports

Generate and download analysis results as multi-sheet Excel workbooks (.xlsx) or CSV files for import into other tools like Power BI or Tableau.

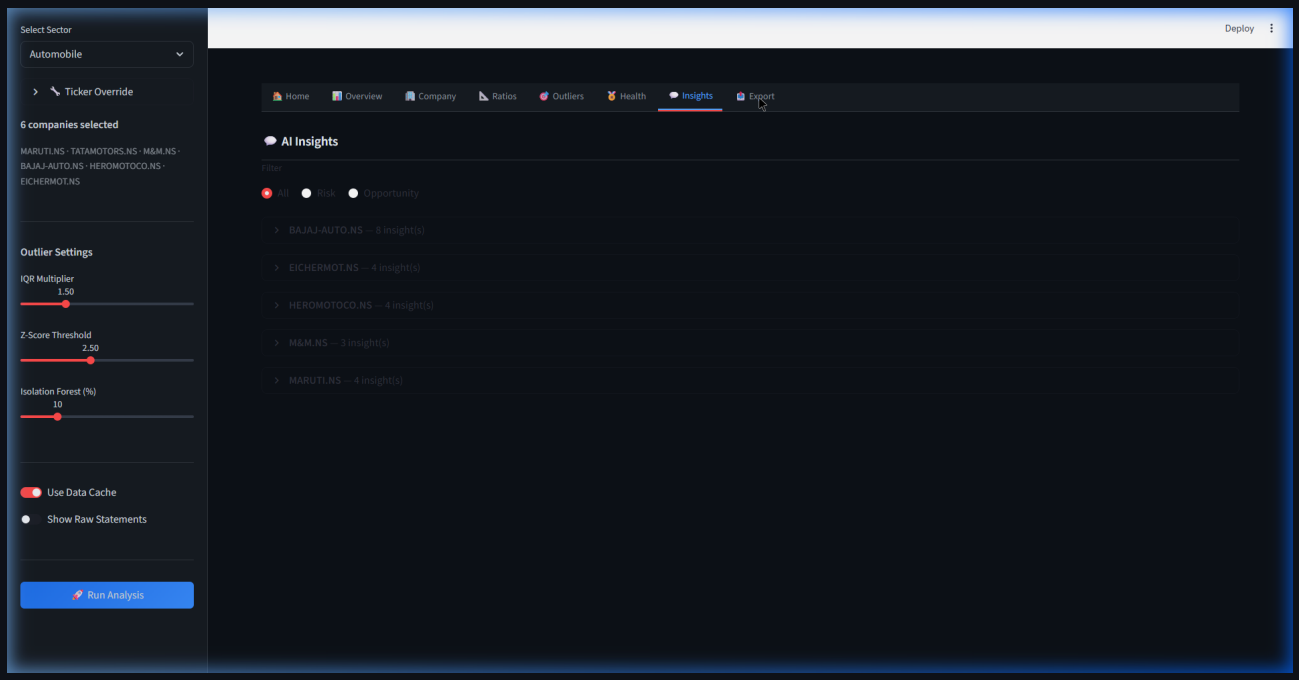


Figure 8 - Export Tab: Download Options

4. How to Use the System

Using the system requires just three steps:

Step 1: Select Sector

Choose from the sidebar dropdown (e.g., Automobile, IT, Banking, Pharma)

Step 2: Run Analysis

Click the 'Run Analysis' button. The system fetches data, computes ratios, and detects outliers automatically (takes 10-30 seconds)

Step 3: Explore Tabs

Navigate through Overview, Company, Ratios, Outliers, Health, and Insights

Optional Customisation

- IQR Multiplier - Controls IQR sensitivity (default: 1.5)
- Z-Score Threshold - How far a value must deviate to be flagged (default: 2.5)
- Isolation Forest % - Expected proportion of anomalies (default: 10%)
- Ticker Override - Manually add or remove companies from the analysis

5. Key Business Benefits

Benefit	Description
Time Savings	Automates hours of data collection - analysis runs in under 30 seconds
Consistency	Same ratios and methods for every company - no human bias
Coverage	Analyses entire sectors at once (5-30+ companies)
Early Detection	Statistical methods catch anomalies missed by traditional screening
Actionable Output	Plain-English insights and exportable reports for decision-making

6. Technical Summary

Component	Technology
Data Source	Yahoo Finance API (yfinance)
Backend	Python 3.13, Pandas, NumPy, SciPy
ML Engine	scikit-learn (Isolation Forest)
Dashboard	Streamlit (web-based, runs locally)
Visualisations	Plotly (interactive charts)
Sectors Covered	13 NSE sectors
Ratios Computed	30+ across 5 pillars
Detection Methods	3 (IQR, Z-Score, Isolation Forest)

To Run

```
streamlit run app.py
```

The dashboard opens in any web browser at http://localhost:8501. No installation beyond Python dependencies is required.

NOTE: This system uses publicly available data from Yahoo Finance. Always cross-reference critical findings with primary sources (NSE filings, annual reports) before making decisions.